Small Errors, Large Residuals: Calibrating PDE-Residual Checks for Neural Surrogates

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Abstract

Neural surrogates for partial differential equations (PDEs) are often declared physically inconsistent when a finite-difference (strong-form) residual of their output exceeds a fixed threshold. We show that such checks conflate how rough the error is, the grid on which the residual is evaluated, and actual violation of the physics. For a linear PDE the strong-form residual of a prediction equals the residual of its error, a symbol-weighted norm whose gain for uncorrelated error grows like h^{-2} and Δt^{-1}, while a weak-form residual is bounded by the relative L^2 error times a roughness-independent constant. We propose reporting either residual as a noise-calibrated ratio C: the residual divided by the median residual of a smooth error of the same relative L^2 on the same grid. On small 1D Burgers and 2D heat datasets, white error at matched relative L^2 has strong-form C of 1.07 to 2.87 (Burgers) and 1.83 to 4.17 (heat), and refining the heat grid from 16 to 48 points per side raises its strong-form residual from 0.875 to 14.4 while the weak-form residual stays between 4.47×10^{-4} and 5.96×10^{-4}. Three shipped FNO and PINO checkpoints that a fixed strong-form cut of 1 flags as untrusted have weak-form C of 0.41 to 0.48, below the smooth-error median, so the verdict does not survive calibration. In a 5-seed study of three architectures, a DeepONet no more accurate than the frozen initial state on Burgers has the lowest weak-form C of the trained models. In a sparse-sensor viscosity inverse on Burgers, a pointwise strong-form estimator fails on 9 of 30 low-viscosity instances and a weak-form estimator on none, and fitting the true solver is more accurate than FNO-conditioned inverses (mean relative error 0.0138 against 0.0337 and 0.0495) at about 10 times the wall clock per instance. Every number comes from committed score files produced on a CPU under frozen protocols.

1 Introduction

Machine learning surrogates such as Fourier neural operators (FNO) [Li et al., 2020], DeepONet [Lu et al., 2021], and physics-informed variants [Raissi et al., 2019, Li et al., 2021] are increasingly judged on whether their outputs are physically consistent, not only on whether they are close to a reference. A common and inexpensive check plugs the prediction into a finite-difference (FD) discretization of the governing equation and compares the norm of what is left over with a threshold. A small residual is read as evidence that the model has learned the physics; a large one as evidence that it has not.

As usually computed, this check answers a different question. For a linear PDE the residual of a prediction is exactly the residual of its error, and the FD operator acts on that error as a

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high-pass filter whose gain grows like h^{-2} for a second-order spatial term and like Δt^{-1} for a first-order time derivative. A fixed threshold therefore mixes the roughness of the error, the evaluation grid, and the physics. We ran into this in an earlier release of our own benchmark, which labeled an FNO "untrusted" with a fixed strong-form cut.[1]

The observation that low prediction error does not imply physically faithful behavior is not new [Takamoto et al., 2022, Wyder et al., 2025, Chakraborty and Chen, 2026]. What we add is an account of why the most common physics-consistency check is hard to interpret, a calibration that fixes the interpretation, and a demonstration that the calibrated verdict can differ from the fixed-threshold one.

Contributions.

1. **Theory (Section 3).** An exact spectral representation of the strong-form FD residual of a prediction in terms of the discrete operator symbol and the error spectrum, an exact second-moment formula for uncorrelated error, and an extension to viscous Burgers by linearization. A weak-form residual is bounded by the L^2 error times a roughness-independent constant and has a white-noise floor that shrinks with resolution.
2. **Calibration protocol (Section 3.5).** The ratio C of a residual to that of a smooth error of the same relative L^2 on the same grid, a 5% to 95% band, and a verdict rule, all in frozen, hashed protocol files.
3. **Controlled evidence (Section 5).** A matched-relL2 ladder of white, smooth, and coefficient-mismatch errors, a resolution sweep, and checks of the second-moment formula (measured/formula 1.00111 for heat and 0.992 for Burgers).
4. **Verdicts change (Section 6).** Shipped FNO and PINO checkpoints flagged by a fixed strong-form cut are not flagged by the calibrated weak-form test.
5. **Model study (Section 7).** Three architectures, 5 seeds each, equal tuning budget, trivial baselines, and paired Wilcoxon tests with effect sizes; a secondary coefficient-response score is in Appendix F.1.
6. **Inverse case study (Section 8).** In sparse-sensor viscosity recovery the pointwise strong-form estimator fails where a weak-form estimator does not, and a solver-in-the-loop fit is more accurate than FNO-conditioned inverses on clean data and far more robust to noise, at higher cost.

Non-claims. Strong-form residuals remain useful as training losses and for smooth errors. A small weak-form residual does not certify a correct solution. The experiments are small (Section 9).

2 Related work

Evaluation practice in scientific ML. McGreivy and Hakim [2024] document weak baselines and reporting biases in ML for fluid-related PDEs. The Common Task Framework (CTF) [Wyder et al., 2025], following Donoho [2017], proposes standardized tasks and naive baselines as the zero point of each score; we adopt that convention for accuracy. PDEBench [Takamoto et al., 2022] reports physics-aware errors; PDEArena [Gupta and Brandstetter, 2022], The Well [Ohana et al., 2024], and CoDBench [Burark et al., 2023] broaden systems and models. None of these analyzes what a pointwise PDE residual measures when used as an evaluation metric.

Scores versus physics. Chakraborty and Chen [2026] show on an airfoil benchmark with paired CFD and PIV data that lower velocity-field error need not coincide with lower error in fluctuation energy. Their analysis is empirical and specific to one benchmark; ours gives an analytic mechanism for one widely used physics check and tests it under controlled error injection.

[1] That verdict was withdrawn in a later release of `https://github.com/shehbaz0101/coticula`; this paper explains why.

Weak forms. Weak SINDy [Messenger and Bortz, 2021a,b] integrates candidate terms against compactly supported test functions to avoid pointwise derivatives of noisy data. Variational PINNs [Kharazmi et al., 2019, 2021], variational operator learning [Xu et al., 2024], and WINO [Zhu et al., 2026] use weak-form residuals as training objectives, and Qiu et al. [2025] build residual objectives provably equivalent to the error in PDE-induced norms. We use the weak form for evaluation and, in Section 8, as the weak-SINDy-style coefficient estimator it was designed to be.

Derivatives of surrogates and architectures. Derivative-informed operator learning matches parametric Jacobians [O'Leary-Roseberry et al., 2022, Yao et al., 2025, Yang et al., 2026]; our coefficient-response score (Appendix F.1) is an evaluation counterpart. We evaluate FNO [Li et al., 2020], a one-level U-Net [Ronneberger et al., 2015], and DeepONet [Lu et al., 2021]; the shipped checkpoints include a PINO [Li et al., 2021].

3 Theory

3.1 Setting and notation

Let $\Omega = [0, \ell)^d$ be periodic, with grid $x_j = jh$, $j \in \{0, \dots, N-1\}^d$, $h = \ell/N$, and time levels $t^n = n\Delta t$. For grid functions we use $\|v\|_h^2 = h^d \sum_j |v_j|^2$ and the discrete Fourier expansion $v_j = \sum_k \hat{v}_k e^{\mathrm{i}k\cdot x_j}$ over the N^d grid wavenumbers, so that Parseval reads $\|v\|_h^2 = \ell^d \sum_k |\hat{v}_k|^2$. The 5-point (in 2D; 3-point in 1D) FD Laplacian acts on Fourier modes as

$$\Delta_h e^{\mathrm{i}k\cdot x} = -\lambda_h(k)\, e^{\mathrm{i}k\cdot x}, \qquad \lambda_h(k) = \sum_{i=1}^{d} \frac{4}{h^2} \sin^2\Big(\frac{k_i h}{2}\Big), \tag{1}$$

with $\frac{4}{\pi^2}|k|^2 \le \lambda_h(k) \le |k|^2$ for $|k_i|\, h \le \pi$ and $\max_k \lambda_h(k) = 4d/h^2$.

The strong-form residual as implemented. We analyze the residual exactly as it is computed in the benchmark code. With midpoint values $v^{n+\frac{1}{2}} = \frac{1}{2}(v^n + v^{n+1})$,

$$\mathcal{R}_h[v]^{n+\frac{1}{2}} = \frac{v^{n+1} - v^n}{\Delta t} - F_h\big(v^{n+\frac{1}{2}}\big), \qquad F_h(v) = \begin{cases} \alpha\, \Delta_h v & \text{(heat)}, \\ -D_0\big(\frac{1}{2}v^2\big) + \nu\, \Delta_h v & \text{(Burgers)}, \end{cases} \tag{2}$$

where $D_0 w_j = (w_{j+1} - w_{j-1})/(2h)$. The reported metric is

$$\rho_{\mathrm{s}}(\hat{u}) = \frac{1}{N_t} \sum_{n=0}^{N_t-1} \frac{\big\|\mathcal{R}_h[\hat{u}]^{n+\frac{1}{2}}\big\|_h}{\big\|\hat{u}^{n+\frac{1}{2}}\big\|_h}. \tag{3}$$

For heat with Dirichlet boundaries the sums run over interior nodes. Write $\hat{u} = u + e$, where u is the reference solution sampled on the grid and e is the prediction error. Since the reference is produced by a different scheme than (2) (or is the continuous solution), $\tau_h := \mathcal{R}_h[u]$ is a nonzero discretization residual; we call $\|\tau_h\|_h \,/\, \|u\|_h$ the *discretization floor*.

3.2 Linear case: the residual is a symbol-weighted norm of the error

Proposition 1 (Exact error identity)**.** *For the heat equation, $\mathcal{R}_h[u+e] = \tau_h + \mathcal{R}_h[e]$ exactly, for any grid functions u, e.*

Proof. $\mathcal{R}_h$ in (2) is linear when $F_h = \alpha\Delta_h$. □

So for a linear PDE the metric sees the reference only through the discretization floor; above the floor, it is a functional of the error alone. The next result describes that functional.

Proposition 2 (Spectral representation)**.** *Let Ω be periodic and $F_h = \alpha\Delta_h$.*

(a) For each n, the Fourier coefficients of $\mathcal{R}_h[e]^{n+\frac{1}{2}}$ are $\dfrac{\hat{e}_k^{n+1} - \hat{e}_k^n}{\Delta t} + \alpha\lambda_h(k)\, \dfrac{\hat{e}_k^{n+1} + \hat{e}_k^n}{2}$.

(b) If in addition e is N_t-periodic in n, with space-time Fourier coefficients $\tilde{e}_{k,\omega}$ over temporal frequencies $\omega \in \frac{2\pi}{N_t \Delta t}\{0, \dots, N_t - 1\}$, then

$$\sum_n \Delta t \left\| \mathcal{R}_h[e]^{n+\frac{1}{2}} \right\|_h^2 = C \sum_{k,\omega} |\sigma_h(k,\omega)|^2 |\tilde{e}_{k,\omega}|^2 ,$$
$$|\sigma_h(k,\omega)|^2 = \frac{4}{\Delta t^2} \sin^2 \frac{\omega \Delta t}{2} + \alpha^2 \lambda_h(k)^2 \cos^2 \frac{\omega \Delta t}{2}, \tag{4}$$

where $C = \ell^d N_t \Delta t$ is the same Parseval constant for which $\sum_n \Delta t \|e^n\|_h^2 = C \sum_{k,\omega} |\tilde{e}_{k,\omega}|^2$.

Proof in Appendix B.

As $h, \Delta t \to 0$ the symbol tends to that of $\partial_t - \alpha\Delta$, unbounded on L^2. Two-sided bounds follow directly: $\min|\sigma_h| \|e\| \le \|\mathcal{R}_h[e]\| \le \max|\sigma_h| \|e\|$ over the support of $\tilde{e}$, with $\max|\sigma_h| = \max(2/\Delta t, 4d\alpha/h^2)$, while a band-limited error ($|k| \le K$, $|\omega| \le \Omega$) has gain at most $\sqrt{\Omega^2 + \alpha^2 K^4}$, independent of h and Δt (Corollary 6, Appendix A).

Effective wavenumber. For a static error the spatial part of the residual reduces to a single scalar, the effective wavenumber $\kappa_h(e) = \left(\sum_k \lambda_h(k)^2 |\hat{e}_k|^2 / \sum_k |\hat{e}_k|^2 \right)^{1/4}$: two static errors of equal norm have residual ratio exactly $(\kappa_h(e_1)/\kappa_h(e_2))^2$ (Corollary 5, Appendix A).

3.3 Uncorrelated error: exact scaling without periodicity

Errors of trained surrogates are not white, but white error is the cleanest probe of the mechanism and is the error model of our injection study. The following holds pointwise and does not require periodic boundaries.

Proposition 3 (White-error second moment, heat)**.** *Let e_j^n be independent with mean zero and variance s^2 at every node and time level. Then at every node whose stencil lies in the grid,*

$$\mathbb{E} \left| \mathcal{R}_h[e]_j^{n+\frac{1}{2}} \right|^2 = s^2 \Big(\frac{2}{\Delta t^2} + \frac{d(2d+1)\,\alpha^2}{h^4} \Big) = \frac{s^2}{\Delta t^2} \big(2 + d(2d+1)\, r^2 \big), \qquad r = \frac{\alpha \Delta t}{h^2}. \tag{5}$$

Proof in Appendix B.

Since $\mathbb{E}\|e^n\|_h^2 = s^2 |\Omega|$, the expected residual-to-error ratio for white error is $C_h = \Delta t^{-1}(2 + d(2d+1)r^2)^{1/2}$. Thus, to leading order in the error and up to the discretization floor and the difference between $\|\hat{u}\|$ and $\|u\|$ in (3),

$$\rho_{\mathrm{s}}(u+e) \approx C_h \cdot \mathrm{relL2}(e), \qquad C_h \ge \max\Big(\frac{\sqrt{2}}{\Delta t},\ \frac{\sqrt{d(2d+1)}\,\alpha}{h^2} \Big), \tag{6}$$

whereas a smooth error band-limited as in Corollary 6(ii) gives at most $\sqrt{\Omega^2 + \alpha^2 K^4} \cdot \mathrm{relL2}(e)$. The parabolic mesh ratio r decides which term dominates: the time difference for $r \ll 1$, the Laplacian for $r \gg 1$. Halving h quadruples the Laplacian contribution; halving Δt doubles the time-difference contribution. For white error $\mathbb{E}\|\Delta_h e\|_h^2 / \mathbb{E}\|e\|_h^2 = 2d(2d+1)/h^4$, so its effective wavenumber is of order $1/h$.

A consequence is an ordering inversion (Corollary 7, Appendix A): the frozen predictor $\hat{u}^n = u_0$ has $\rho_{\mathrm{s}} = \alpha \kappa_h(u_0)^2$ regardless of how far the solution moves, so a slightly noisy exact solution can score worse than a predictor that ignores the dynamics.

Viscous Burgers. The residual splits exactly into τ_h, a linearized term, and a quadratic term; for i.i.d. Gaussian error the pointwise second moment is $s^2\big(2/\Delta t^2 + 3\nu^2/h^4 + O(u^2/h^2)\big) + s^4/(16h^2)$ (Proposition 8, Appendix A).

3.4 The weak-form residual

Let $\{\phi_m\}_{m=1}^M$ be smooth test functions on $\Omega \times [0,T]$ that vanish, with their time derivatives, near $t=0$ and $t=T$, are periodic in x for Burgers, and vanish on $\partial\Omega$ for heat with Dirichlet boundaries. Define, with no derivative on v,

$$r_m[v] = -\int_0^T \int_\Omega v\left(\partial_t \phi_m + \alpha \Delta \phi_m\right) dx\, dt = \langle v,\, L^* \phi_m \rangle\,, \qquad L^* = -\partial_t - \alpha\Delta \quad \text{(heat)}, \tag{7}$$

$$r_m[v] = -\int_0^T \int_\Omega \Big(v\, \partial_t \phi_m + \tfrac{1}{2} v^2\, \partial_x \phi_m + \nu\, v\, \partial_{xx} \phi_m\Big) dx\, dt \quad \text{(Burgers)}. \tag{8}$$

Integration by parts shows $r_m[u] = 0$ for every classical solution u (for heat, using $u = \phi_m = 0$ on $\partial\Omega$). In practice r_m is computed by quadrature of $\hat{u}$ against analytically differentiated ϕ_m, as in weak SINDy [Messenger and Bortz, 2021b], so the quadrature error on u plays the role of τ_h. Let $\psi_m = L^* \phi_m$, $G_{mn} = \langle \psi_m, \psi_n \rangle$, and define

$$\rho_{\mathrm{w}}(\hat{u}) = \frac{\left(\sum_m r_m[\hat{u}]^2\right)^{1/2}}{\|u\|\, \left(\operatorname{tr} G\right)^{1/2}}. \tag{9}$$

For Burgers we use the same normalization with the linear part $\psi_m = -\partial_t \phi_m - \nu \partial_{xx} \phi_m$; the residual r_m itself includes the convective term.

Proposition 4 (Roughness-independent bound, heat)**.** *Let u be a classical solution and $\hat{u} = u + e$ with $e \in L^2$. Then $r_m[\hat{u}] = \langle e, \psi_m \rangle$ and*

$$\rho_{\mathrm{w}}(\hat{u}) \le \left(\frac{\lambda_{\max}(G)}{\operatorname{tr} G}\right)^{1/2} \frac{\|e\|}{\|u\|} \le \frac{\|e\|}{\|u\|}. \tag{10}$$

The bound depends on e only through $\|e\|$.

Proof in Appendix B.

For the strong form the corresponding constant is $\max(2/\Delta t, 4d\alpha/h^2)$; here it is at most one. For i.i.d. error with equal quadrature weights on N_{pts} nodes the expected weak residual is exactly $1/\sqrt{N_{\mathrm{pts}}}$ of the bound (Proposition 9, Appendix A), while a smooth error in the span of the ψ_m attains it: the weak residual inverts the roughness preference of the strong residual. Detecting high-frequency error is then the job of the accuracy metric.

Conversely, the weak residual does respond to physics: a classical solution with the wrong diffusivity α' gives $r_m = (\alpha' - \alpha)\, \langle \hat{u},\, \Delta\phi_m \rangle$, and the frozen predictor gives $r_m = -\alpha\, \langle u_0,\, \Delta\phi_m \rangle$ (Proposition 10); for Burgers the weak residual is first order in $\|e\|$ with a roughness-independent constant plus a quadratic term.

3.5 The noise-calibrated ratio

Propositions 3, 4, and 9 show that a raw residual value has no meaning without the error size and the grid it was computed on. We therefore report residuals relative to a generic smooth error of the same size on the same data and grid. For a field $\hat{u}$ with $\mathrm{relL2}(\hat{u}) = \varepsilon$ and a residual metric $\rho \in \{\rho_{\mathrm{s}}, \rho_{\mathrm{w}}\}$,

$$C(\hat{u}) = \frac{\rho(\hat{u})}{\operatorname{med}\, \rho\big(u + e^{(\varepsilon)}_{\mathrm{smooth}}\big)}, \tag{11}$$

where $e^{(\varepsilon)}_{\mathrm{smooth}}$ is a time-varying smooth low-mode error rescaled to relative L^2 equal to ε, and the median is over trajectories and random draws. $C > 1$ means less PDE-consistent than a generic smooth error of the same size; $C < 1$ means more. The 5% to 95% quantiles of the same distribution give a band; a field is flagged only if its C lies above the band. The choice of the smooth null is a modeling decision, discussed in Section 9.

4 Experimental setup

Protocols. All choices are fixed in `protocol.json` (v1, SHA-256 prefix `8b9110746356`), frozen before any test scoring, and in an amendment `protocol_v2.json` (prefix

Table 1: Matched-relL2 ladder on exact solver outputs (protocol v2). Smooth ρ is the median residual of a time-varying smooth error at that ε; $C = \rho/\rho_{\text{smooth}}$ for white noise and for the amplitude-matched coefficient-mismatch error (2× coefficient). $C > 1$: less PDE-consistent than a generic smooth error of the same size. Medians over trajectories and draws. Source: `v2_matched_ladder.csv`.

PDE	ε	smooth ρ_{s}	white C_s	mism. C_s	smooth ρ_{w}	white C_w	mism. C_w
Burgers 1D	1.00×10^{-3}	0.0569	1.07	0.966	2.59×10^{-4}	0.999	0.940
	0.0100	0.0981	2.38	0.434	6.39×10^{-4}	0.777	0.279
	0.0500	0.401	2.82	0.205	2.95×10^{-3}	0.726	0.152
	0.100	0.794	2.84	0.235	5.88×10^{-3}	0.733	0.170
	0.200	1.57	2.84	0.272	0.0120	0.707	0.177
	0.500	3.72	2.87	0.355	0.0299	0.719	0.181
Heat 2D	1.00×10^{-3}	0.0939	1.83	0.875	4.65×10^{-4}	0.990	1.00
	0.0100	0.367	3.62	0.328	7.27×10^{-4}	0.691	0.762
	0.0500	1.72	3.83	0.195	2.77×10^{-3}	0.421	0.394
	0.100	3.35	3.91	0.193	5.51×10^{-3}	0.392	0.338
	0.200	6.49	3.97	0.219	0.0107	0.406	0.322
	0.500	13.6	4.17	0.736	0.0279	0.384	0.297

`6457f447ab55`) written after the v1 results were available and frozen before v2 scoring. It keeps data, models, and the weak form and changes the analysis (score S only for relL2, residuals as C, smooth-band verdict null, Wilcoxon tests; Appendix C). The v2 analyses are therefore not strictly pre-registered; v1 outcomes are reported alongside.

PDEs and data. *1D viscous Burgers* on a periodic interval of length 2π, 32 grid points, 16 time steps, $\nu \in [0.005, 0.05]$, 20 trajectories. *2D heat* on the unit square with homogeneous Dirichlet boundaries, 16×16 grid, 12 time steps, $\alpha \in [0.05, 0.2]$, 16 trajectories. Each PDE has 4 test trajectories (Table 6). Shipped checkpoints are evaluated on their original datasets (Burgers 64 points and 80 steps; heat 32×32 and 40 steps).

Residuals. ρ_{s} is Eq. (3); ρ_{w} is Eq. (9) with test functions $\chi_b(t)\,\eta_a(x)$: degree-8 polynomial bumps in time [Messenger and Bortz, 2021b] times Fourier modes $k \leq 3$ (Burgers) or Dirichlet sine modes $p, q \leq 3$ (heat), so no boundary term appears.

Injected errors. For each test trajectory and level $\varepsilon \in \{0.001, 0.01, 0.05, 0.1, 0.2, 0.5\}$ the error is rescaled so that relL2 $= \varepsilon$ exactly. *White*: i.i.d. Gaussian at every node and time (for heat, set to zero on the boundary before rescaling). *Smooth*: each spatial mode $1, \ldots, 4$ has an independent trigonometric polynomial of degree 2 in time as its coefficient, so the error varies in time. *Coefficient mismatch*: $u(2c) - u(c)$ from the same initial state, rescaled to ε and added to $u(c)$; residuals use the original coefficient c. The ladder and bands use 16 draws per trajectory (8 for shipped checkpoints).

Models and statistics. FNO, U-Net, and DeepONet are trained with data MSE only under the same budget: 2 configurations of 40 epochs selected on validation relL2, then 120 epochs with 5 seeds (Table 7). Trivial baselines are zeros, the training-mean field, and the frozen initial state. $S(M) = 100\,(M_0 - M)/(M_0 - M_{\text{ref}})$ with $M_{\text{ref}} = 0$ and M_0 the best trivial baseline, not clipped. Contrasts use two-sided Wilcoxon signed-rank tests over $n = 20$ (seed, test trajectory) pairs with the mean paired difference and Cohen's d_z. Both reproduction runs together take 77 s on a CPU (Table 8).

5 Noise injection, resolution, and coefficient mismatch

The strong form ranks roughness. At every level the strong-form residual of white error exceeds that of smooth error of the same size: white C_s ranges from 1.07 to 2.87 on Burgers and from 1.83 to 4.17 on heat (Table 1; Figure 1 in Appendix F); the pre-registered paired test at $\varepsilon = 0.01$ gives $p = 5.0 \times 10^{-4}$ on both PDEs, the smallest value attainable with

Table 2: Shipped checkpoints (on their original evaluation sets) and the frozen-initial baseline. Legacy flag: $\rho_s > 1$. v1 rule: weak residual above the matched white-noise envelope. v2 rule: weak C above the 95% smooth band at the field's own relL2; not called outside the noise grid. Sources: `v2_shipped_survival.csv`, `shipped_rescore.csv` (v1 column).

PDE	model	relL2	ρ_s	legacy flag	C_s	C_w	v1 survives	v2 survives
Burgers 1D	FNO	0.157	4.85	yes	3.85	0.410	yes	no
Burgers 1D	frozen initial	0.377	0.809	no	0.280	0.235	–	–
Burgers 1D	PINO	0.188	1.73	yes	1.15	0.443	yes	no
Heat 2D	FNO	0.322	13.0	yes	1.37	0.484	yes	no
Heat 2D	frozen initial	1.42	12.8	yes	–	–	yes	not called
Heat 2D	PINO	checkpoint not available (missing checkpoint)						

2000 sign-flip permutations (Table 20). The separation is modest because these grids are coarse. Corollary 5 bounds the spatial part of the ratio by $(\kappa_h(e_{\text{white}})/\kappa_h(e_{\text{smooth}}))^2$, which is 7.1 for Burgers and 6.3 for heat on these grids (Table 12). Which term of Proposition 3 dominates depends on $r = c\,\Delta t/h^2$. On the Burgers grid r lies between 8.1×10^{-3} and 0.081, so the spatial term is at most 9.9×10^{-3} times the time-difference term and the time difference dominates; the smooth error also varies in time over only 16 steps, so the temporal separation is small too. On the heat grid r lies between 0.47 and 1.9, so the spatial term is 1.1 to 18 times the time term: comparable at the low end of α and dominant at the high end. At the smallest level the residuals approach the discretization floor, and on Burgers C approaches 1.

The gap grows with resolution. At $\varepsilon = 0.01$, refining heat from 16 to 48 points per side raises the white-error ρ_s from 0.875 to 14.4 while the smooth-error ρ_s goes from 0.265 to 0.643, so the white/smooth ratio grows from 3.30 to 22.4; the mesh ratio grows from 0.82 to 12.1 and the spatial term dominates more and more (3.4 to 734 times the time term). The white-error ρ_w goes from 5.96×10^{-4} to 4.47×10^{-4} (Figure 2 in Appendix F). On Burgers, where the time term dominates, white ρ_s rises from 0.283 to 0.458 as the grid is refined and from 0.329 to 0.570 as Δt is reduced, while white ρ_w decreases from 4.32×10^{-4} to 2.22×10^{-4}. The second-moment prediction matches the measured white ρ_s to a factor between 0.994 and 2.21 on Burgers and between 1.23 and 1.90 on heat (Table 11); the theory ignores the discretization floor, and on heat it underpredicts at every grid, which we have not fully explained. On synthetic i.i.d. fields the measured/formula ratio is 1.00111 (heat) and 0.992 (Burgers), the static identity of Corollary 5 holds to 1.000000, and all 10 of 10 pre-registered checks are supported (Tables 19 and 20).

The weak form does not reward smoothness of the error. White C_w is below 1 at every level, from 0.707 to 0.999 on Burgers and from 0.384 to 0.990 on heat, as Proposition 9 predicts.

Coefficient errors look more physical than generic smooth errors. For $\varepsilon \geq 0.01$ the amplitude-matched mismatch error has strong-form C_s between 0.20 and 0.43 (Burgers) and 0.19 and 0.74 (heat), and weak-form C_w between 0.15 and 0.28 (Burgers) and 0.30 and 0.76 (heat). The heat weak-form case at $\varepsilon = 0.01$ ($C_w = 0.76$) is close to the smooth median; at $\varepsilon = 0.001$ the values are near 1 (0.94 and 1.0), next to the discretization floor. So at matched relL2 neither residual singles out a coefficient error: a solution of a nearby PDE is a smooth, solution-like perturbation, and both forms rate it as more consistent than a generic smooth error. Scoring exact labels with a zero coefficient does raise ρ_w by a factor of 3.02 (Burgers) and 27.2 (heat) over the true coefficient (Table 17), in line with Proposition 10. Wrong coefficient responses are better probed directly (Appendix F.1).

6 Calibration changes the verdict on shipped checkpoints

The legacy rule flags the shipped Burgers FNO ($\rho_s = 4.85$), Burgers PINO (1.73), and heat FNO (13.0) as untrusted, and passes the Burgers frozen initial state ($\rho_s = 0.809$) although its relL2 is 0.377 (Table 2). Calibrated, the strong-form ratios are $C_s = 3.85$, 1.15, and 1.37: the Burgers FNO error is rougher than a smooth error of the same size, the other two are

Table 3: Model study (protocol v2): three architectures, mean ± std over 5 seeds; trivial baselines are deterministic. S(relL2): best trivial baseline = 0, reference = 100, not clipped. C is undefined (–) when relL2 lies outside the calibrated noise grid. Source: `v2_calibrated_ratios.csv`.

PDE	model	relL2	S(relL2)	ρ_s	C_s	ρ_w	C_w
Burgers 1D	zeros	1.000	−169	0	–	0	–
	training mean	0.997	−169	1.01	–	8.43×10^{-4}	–
	frozen initial	0.371	0	0.748	0.266 ± 0.012	4.84×10^{-3}	0.220 ± 0.056
	FNO	0.0744 ± 0.0022	80.0	0.450	0.769 ± 0.048	1.67×10^{-3}	0.379 ± 0.046
	U-Net	0.108 ± 0.006	70.9	1.03	1.20 ± 0.18	3.66×10^{-3}	0.571 ± 0.204
	DeepONet	0.377 ± 0.011	−1.52	0.906	0.341 ± 0.019	2.46×10^{-3}	0.137 ± 0.022
	reference	0	100	0.0539	–	2.79×10^{-4}	–
Heat 2D	zeros	1.000	−100	0	–	0	–
	training mean	0.499	0	0.961	0.0857 ± 0.0378	2.74×10^{-3}	0.121 ± 0.070
	frozen initial	1.39	−178	11.5	–	0.0512	–
	FNO	0.183 ± 0.002	63.4	4.89	0.837 ± 0.041	5.57×10^{-3}	0.552 ± 0.059
	U-Net	0.138 ± 0.007	72.3	4.61	1.05 ± 0.08	5.45×10^{-3}	0.718 ± 0.063
	DeepONet	0.308 ± 0.023	38.4	8.15	0.878 ± 0.152	8.11×10^{-3}	0.478 ± 0.029
	reference	0	100	0.0940	–	4.70×10^{-4}	–

close to it. The weak-form ratios are $C_w = 0.410$, 0.443, and 0.484, below 1 and so below both the smooth median and the 95% edge of the band; none of the three untrusted verdicts survives the v2 rule. Under the v1 rule, whose null is the white-noise envelope, all three would survive; we replaced that rule because clearing a $1/\sqrt{N}$ floor says little (Section 4). The heat frozen state lies outside the calibrated relL2 range and is not called; the heat PINO checkpoint was not available. Read together, what the fixed cut detected on the Burgers FNO is mainly the roughness of its error, and on the other two little that a generic smooth error of the same size would not also produce.

7 Model study

Accuracy. On Burgers, FNO is the most accurate (S(relL2) = 80.0); on heat, U-Net is (FNO minus U-Net relL2: 0.0441, $d_z = 3.37$, $p = 1.9 \times 10^{-6}$; Table 13). The Burgers DeepONet does not beat the frozen initial state ($S = -1.5$).

Low residual is not accuracy. The same Burgers DeepONet has the lowest weak-form ratio of the trained models ($C_w = 0.14$; FNO minus DeepONet C_w: 0.248, $d_z = 2.50$, $p = 1.9 \times 10^{-6}$), and on heat the training-mean field has $C_w = 0.12$, below every trained model. A predictor that moves little or produces smooth fields scores well on any residual.

Calibration separates what raw residuals do not. On heat the raw ρ_s of FNO and U-Net does not differ significantly (difference 0.280, $p = 0.083$), but C_s does (−0.218, $d_z = -1.26$, $p = 2.7 \times 10^{-5}$), because C accounts for the larger error of the FNO. FNO and DeepONet on heat are indistinguishable in C_s ($p = 0.90$). Raw and calibrated residuals can rank models differently, and neither ranking agrees with accuracy (v1 residual scores: Table 14).

8 Inverse case study: viscosity from sparse sensors

The same mechanism decides which residual is safe to fit when it becomes an estimator. On a separate Burgers dataset (PINNForge, Appendix E) we recover ν from 32 equispaced sensors and 4 bursts of 5 frames and score the low-viscosity `hard_ood` slice under a protocol frozen before test scoring (SHA-256 prefix `055aecf6ab7c`). Least squares on the pointwise strong-form residual at the sensors fails worst (mean relative error 0.349, 9 failures; Table 4): on a 32-point grid u_{xx} of a steep low-ν solution is aliased, so the fit absorbs resolution error as physics. A weak-form estimator in the style of weak SINDy, which moves every derivative onto smooth test functions, removes all failures (0.0898, 0), the inverse analogue of

Table 4: Burgers viscosity recovery on `hard_ood` ($n = 30$, $\nu \leq 0.0270$) from 32 sensors and 4 bursts, clean data. Relative error of $\hat{\nu}$; failure: $\hat{\nu} \leq 0$ or relative error > 0.5. FNO rows: mean $\pm$ std over 5 seeds of committed per-instance estimates. Last two columns: mean paired difference FNO minus estimator, and in parentheses the number of instances (mean over seeds) on which the FNO is lower. Seconds per instance on CPU; n/m: not measured; †FNO timing uses random weights of the same architecture. Source: `strong_baselines_scores.json`.

estimator	mean	median	max	fail.	s/inst.	Δ data-only (wins)	Δ hybrid (wins)
Strong-form residual LS (closed form)	0.349	0.165	0.950	9	n/m	-0.300 (29.4)	-0.316 (29.2)
Weak-form LS	0.0898	0.0524	0.495	0	1.28×10^{-3}	-0.0403 (24)	-0.0561 (26.2)
Solver fit, sensor-lift IC	0.0138	9.38×10^{-3}	0.0551	0	18.7	0.0357 (4.4)	0.0199 (7.6)
Solver fit, oracle IC (ceiling)	2.51×10^{-3}	2.28×10^{-3}	6.99×10^{-3}	0	16.5	0.0470 (1.6)	0.0312 (2.8)
FNO inverse, data-only	0.0495 ± 0.0144	0.0315 ± 0.0156	0.292 ± 0.054	0	$1.85^{\dagger}$	–	–
FNO inverse, hybrid 0.01	0.0337 ± 0.0035	0.0191 ± 0.0026	0.187 ± 0.054	0	$1.85^{\dagger}$	–	–

Sections 5 and 6. Adding that aliased residual to the FNO objective also hurts: on validation `hard_ood` the data-only arm goes from 0.0453 to 0.118 and 0.268 as the residual weight rises from 0 to 1 and 10. Both FNO-conditioned inverses beat both residual estimators (0.0495 and 0.0337; lower than the weak form on 24 and 26.2 of 30 instances), but fitting the true spectral solver to the sensors beats both FNO arms (0.0138; the FNO is lower on only 4.4 and 7.6 instances), with an oracle-initial-state ceiling of 2.51×10^{-3}, at about 10 times the wall clock (18.7 s against 1.85 s per instance). Under Gaussian noise (Table 9) the FNO arms reach 0.179 and 0.183 at 2% and 0.750 and 0.835 at 5%, where they are worse than the closed form (0.299) and the weak form (0.591), while the solver stays at 0.0182. With 16 sensors the sensor-lift solver (0.154, 2 failures) and the FNO arms (0.175 and 0.161, 1 failure each) are close, and the weak form collapses (1.81) because its fixed radius of 8 sensor spacings now spans the whole domain; with 8 sensors every method fails. A surrogate can beat residual-based identification without beating the solver it imitates.

9 Limitations

- **Small scale.** Two PDEs on coarse grids (32 points; 16×16), 4 test trajectories per PDE, one dataset seed, small CPU-trained models with 2 tuning trials.
- **The null is a choice.** C uses a specific smooth error (modes up to 4, temporal degree 2); another null changes C and the band.
- **The weak residual is not a certificate.** A finite test space ($K_\phi = 3$; ablation in Table 18) has a null space that contains most high-frequency errors, and the zero field has zero residual. Neither residual singles out coefficient errors at matched relL2.
- **Post hoc amendment.** The v2 rule, C, and the Wilcoxon tests were fixed after the v1 results were seen; under the v1 rule the shipped verdicts survive. The heat PINO checkpoint was unavailable.
- **Theory scope.** Exact results need linear operators on periodic grids or uncorrelated error; Burgers is linearized. Heat residuals exceed the second-moment prediction (Section 5).
- **Inverse study scope.** One scalar parameter on 1D periodic Burgers. Weak-form radii were not tuned. FNO errors on clean data are the committed Stage D per-instance estimates, not retrained (stress FNO numbers: committed Stage E retrain). The solver uses a 7-point coarse grid plus at most 10 golden-section steps, the FNO a 191-point grid; the solver was stressed on 4 cells only.

10 Conclusion

A fixed threshold on a strong-form FD residual mixes the roughness of the error, the grid, and the physics. Calibrating against a smooth error of the same size, with a weak form, changes verdicts on shipped checkpoints, exposes low residuals of inaccurate models, and in a viscosity inverse separates a failing pointwise estimator from a reliable weak-form one. Claims of physics consistency should report accuracy, a calibrated weak-form residual, the null, and the grid.

Reproducibility statement

Sections 5 to 7: `https://github.com/shehbaz0101/coticula`, branch `cursor/weak-form-residual-study-6dd7`, pull request #8, commit `8a9acf9`; `scripts/reproduce_paper.sh` regenerates `paper_artifacts/` on a CPU. Section 8: `https://github.com/shehbaz0101/pinnforge`, branch `cursor/strong-baselines-viscosity-bc36`, pull request #24, commit `7d90f6f`; scores in `docs/research/strong_baselines_scores.json` and `docs/v02/`. `tools/make_tables.py` in the paper source turns these committed files into every table and number. Frozen protocol SHA-256 hashes:

coticula v1: `8b9110746356564fb3cf1ac6e8cd1b6cd223c4b022b366e99c93c85c5520d477`
coticula v2: `6457f447ab555e30803068fd92d7d36738d293d09616a9b4174bd0300c990549`
PINNForge strong baselines: `055aecf6ab7c3611c78652c8ddf727fe591158f51c670cb22f0556fddc129a86`

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A Additional theory

Effective wavenumber.

Definition 1. For a nonzero grid function e on Ω, the effective wavenumber is $\kappa_h(e) = \big(\sum_k \lambda_h(k)^2 \,|\hat{e}_k|^2 \,/\, \sum_k |\hat{e}_k|^2\big)^{1/4}$.

By (1), $\frac{2}{\pi} k_4(e) \le \kappa_h(e) \le k_4(e)$, where $k_4(e) = (\sum_k |k|^4 \,|\hat{e}_k|^2 \,/\, \sum_k |\hat{e}_k|^2)^{1/4}$ is the energy-weighted fourth-moment wavenumber of e.

Corollary 5 (Static error; matched-L^2 ratio)**.** *If the error does not change in time, $e^n = e$ for all n, then $\mathcal{R}_h[e] = -\alpha \Delta_h e$ and*

$$\frac{\|\mathcal{R}_h[e]\|_h}{\|u\|_h} = \alpha\, \kappa_h(e)^2\, \frac{\|e\|_h}{\|u\|_h}. \tag{12}$$

For two static errors with $\|e_1\|_h = \|e_2\|_h$, the residual ratio is exactly $(\kappa_h(e_1)/\kappa_h(e_2))^2$. If e_2 is band-limited to $|k| \le K$ then $\kappa_h(e_2) \le K$.

Proof in Appendix B.

Corollary 6 (Bounds)**.** *Under the assumptions of Proposition 2(b), let S be the set of (k, ω) on which $\tilde{e}$ is supported. Then $\min_S |\sigma_h| \, \|e\| \le \|\mathcal{R}_h[e]\| \le \max_S |\sigma_h| \, \|e\|$ in the space-time norm of (4). In particular:*

(i) for any e, $\|\mathcal{R}_h[e]\| \le \max\big(2/\Delta t,\ 4d\alpha/h^2\big)\, \|e\|$;

(ii) if e is band-limited, $|k| \le K$ and $|\omega| \le \Omega$ with $\Omega \Delta t \le \pi$, then $\|\mathcal{R}_h[e]\| \le \sqrt{\Omega^2 + \alpha^2 K^4}\, \|e\|$, independent of h and Δt.

Proof in Appendix B.

Corollary 7 (Ordering inversion)**.** *Let u_0 be the initial state and consider the frozen predictor $\hat{u}^n = u_0$ for all n. Its residual is $\mathcal{R}_h[\hat{u}] = -\alpha \Delta_h u_0$, so $\rho_{\mathrm{s}}(\hat{u}) = \alpha \kappa_h(u_0)^2$, which is small for a smooth initial state and does not depend on how far the true solution moves. A white-noise perturbation of the exact solution has $\rho_{\mathrm{s}} \approx C_h \,\mathrm{relL2}$. Hence, whenever $\mathrm{relL2} > \alpha \kappa_h(u_0)^2 / C_h$ (up to the discretization floor), the strong-form metric ranks the noisy exact solution as less physical than a predictor that ignores the dynamics.*

A.1 Viscous Burgers

Proposition 8 (Linearization)**.** *For F_h in (2) (Burgers), with all spatial operators applied at the midpoint,*

$$\mathcal{R}_h[u+e] = \tau_h + \mathcal{L}_{u,h} e + \mathcal{Q}_h(e), \quad \mathcal{L}_{u,h} e = \frac{e^{n+1} - e^n}{\Delta t} + D_0\big(u\, e\big) - \nu \Delta_h e, \quad \mathcal{Q}_h(e) = D_0\big(\tfrac{1}{2} e^2\big), \tag{13}$$

exactly. If e_j^n are i.i.d. $\mathcal{N}(0, s^2)$, then at node j (with $u = u^{n+\frac{1}{2}}$)

$$\mathbb{E}\,|\mathcal{R}_h[u+e]_j - \tau_{h,j}|^2 = s^2 \Big(\frac{2}{\Delta t^2} + \frac{3\nu^2}{h^4} + \frac{u_{j+1}^2 + u_{j-1}^2}{8h^2} - \frac{\nu\,(D_0 u)_j}{h^2} \Big) + \frac{s^4}{16 h^2}. \tag{14}$$

Proof in Appendix B.

With frozen coefficient $u \equiv \bar{u}$ the linear part has symbol $e^{\mathrm{i}\theta/2}\big[\frac{2\mathrm{i}}{\Delta t} \sin\frac{\theta}{2} + \big(\nu \lambda_h(k) + \mathrm{i}\,\bar{u} \frac{\sin kh}{h}\big) \cos\frac{\theta}{2}\big]$: the convective term adds a first-order contribution of size $|\bar{u}|\,/h$ at high wavenumber, subdominant to the viscous ν/h^2 term when the cell Péclet number $|\bar{u}|\, h/\nu$ is small. For small ν the time-difference term $2/\Delta t^2$ dominates (14) unless $3r^2/2$ is of order one, with $r = \nu \Delta t / h^2$.

Proposition 9 (White-noise floor)**.** *Let the quadrature have equal weights $w = h^d \Delta t$ on N_{pts} nodes, $|D| = N_{\mathrm{pts}} w$, and let e be i.i.d. with mean zero and variance s^2. Then $\mathbb{E} \sum_m r_m[e]^2 = s^2 w \,\mathrm{tr}\, G_h$ and $\mathbb{E}\,\|e\|_h^2 = s^2\,|D|$, where G_h is the quadrature Gram matrix. Hence*

$$\frac{\mathbb{E} \sum_m r_m[e]^2}{\mathrm{tr}\, G_h\ \mathbb{E}\,\|e\|_h^2} = \frac{1}{N_{\mathrm{pts}}}. \tag{15}$$

Proof in Appendix B.

Proposition 10 (Physics mismatch is detected)**.** *For heat with test functions as above:*

(a) if $\hat{u}$ is a classical solution with the wrong diffusivity α' and zero boundary values, then $r_m[\hat{u}] = (\alpha' - \alpha)\,\langle \hat{u},\, \Delta\phi_m \rangle$;

(b) for the frozen predictor $\hat{u}(x,t) = u_0(x)$, $r_m[\hat{u}] = -\alpha\,\langle u_0,\, \Delta\phi_m \rangle$.

Proof in Appendix B.

Both signals scale with $\langle \hat{u},\, \Delta\phi_m \rangle$, a smooth functional of the prediction, rather than with the noise level. For Burgers, $r_m[u+e] = -\langle e,\, \partial_t\phi_m + u\,\partial_x\phi_m + \nu\partial_{xx}\phi_m \rangle - \frac{1}{2}\langle e^2,\, \partial_x\phi_m \rangle$, so $|r_m[u+e]| \le \|e\|\,\|L_u^*\phi_m\| + \frac{1}{2}\|e\|^2\,\|\partial_x\phi_m\|_\infty$, with $L_u^*\phi = -\partial_t\phi - u\partial_x\phi - \nu\partial_{xx}\phi$: again first order in $\|e\|$ with a roughness-independent constant, plus a quadratic term.

B Deferred proofs

Proof of Proposition 2. (a) follows from (1) and linearity. For (b), insert the mode $e_j^n = e^{\mathrm{i}(k\cdot x_j + \omega t^n)}$ and write $\theta = \omega\Delta t$. Then $\mathcal{R}_h[e]_j^{n+\frac{1}{2}} = \sigma_h e_j^n$ with $\sigma_h = (e^{\mathrm{i}\theta} - 1)/\Delta t + \alpha\lambda_h(1 + e^{\mathrm{i}\theta})/2 = e^{\mathrm{i}\theta/2}\big(\frac{2\mathrm{i}}{\Delta t}\sin\frac{\theta}{2} + \alpha\lambda_h\cos\frac{\theta}{2}\big)$. The two terms in the bracket are imaginary and real, which gives (4). Space-time Fourier modes are orthogonal under periodicity, so Parseval applies mode by mode. □

Proof of Corollary 5. The time difference vanishes and the midpoint average equals e. Then $\|\Delta_h e\|_h^2 = \ell^d \sum_k \lambda_h^2\,|\hat{e}_k|^2 = \kappa_h(e)^4\,\|e\|_h^2$ by Parseval. □

Proof of Proposition 3. $\mathcal{R}_h[e]_j^{n+\frac{1}{2}}$ is a linear combination of independent variables. The coefficient of e_j^{n+1} is $1/\Delta t + d\alpha/h^2$, that of e_j^n is $-1/\Delta t + d\alpha/h^2$, and each of the $2d$ neighbors at each of the two time levels has coefficient $-\alpha/(2h^2)$. The sum of squared coefficients is $2/\Delta t^2 + 2d^2\alpha^2/h^4 + 4d\cdot\alpha^2/(4h^4)$. □

Proof of Proposition 4. Linearity and $r_m[u] = 0$ give $r_m[\hat{u}] = \langle e,\, \psi_m \rangle$. Then $\sum_m \langle e,\, \psi_m \rangle^2 = \sup_{\|a\|=1} \langle e,\, \sum_m a_m\psi_m \rangle^2 \le \|e\|^2\,\lambda_{\max}(G)$, and $\lambda_{\max}(G) \le \operatorname{tr} G$. □

Proof of Corollary 6. The two-sided bound is immediate from (4). For (i), $A\sin^2 + B\cos^2 \le \max(A,B)$ with $\lambda_h \le 4d/h^2$. For (ii), $|\sin(\theta/2)| \le |\theta|/2 \le \Omega\Delta t/2$, $\cos^2 \le 1$, and $\lambda_h(k) \le |k|^2 \le K^2$. □

Proof of Proposition 8. The decomposition follows from $(u+e)^2 = u^2 + 2ue + e^2$ and linearity of D_0 and Δ_h. For the second moment, the coefficients of e_j^{n+1} and e_j^n in $\mathcal{L}_{u,h}e$ are $\pm 1/\Delta t + \nu/h^2$; those of e_{j+1}^n and e_{j+1}^{n+1} are $u_{j+1}/(4h) - \nu/(2h^2)$; those of e_{j-1}^n and e_{j-1}^{n+1} are $-u_{j-1}/(4h) - \nu/(2h^2)$. Summing squares gives the s^2 term. The midpoint error $e^{n+\frac{1}{2}}$ is $\mathcal{N}(0, s^2/2)$ and independent across nodes, so $\mathcal{Q}_h(e)_j = \big((e_{j+1}^{n+\frac{1}{2}})^2 - (e_{j-1}^{n+\frac{1}{2}})^2\big)/(4h)$ has mean zero and variance $2\cdot\frac{s^4}{2}/(16h^2)$. Its cross moment with $\mathcal{L}_{u,h}e$ involves only odd moments of a centered Gaussian and vanishes. □

Proof of Proposition 9. $r_m[e] = w\sum_i e_i\psi_{m,i}$, so $\mathbb{E}r_m[e]^2 = w^2s^2\sum_i \psi_{m,i}^2 = ws^2\,\|\psi_m\|_h^2$. □

Proof of Proposition 10. (a) $-\langle \hat{u},\, \partial_t\phi_m \rangle = \langle \partial_t\hat{u},\, \phi_m \rangle = \alpha'\,\langle \Delta\hat{u},\, \phi_m \rangle = \alpha'\,\langle \hat{u},\, \Delta\phi_m \rangle$, using the vanishing of ϕ_m near $t \in \{0, T\}$ and on $\partial\Omega$. (b) $\int_0^T \partial_t\phi_m\,dt = 0$. □

Table 5: SHA-256 of the frozen protocol files (from `protocol.sha256`, `protocol_v2.sha256`).

file	SHA-256
`protocol.json` (v1)	`8b9110746356564fb3cf1ac6e8cd1b6cd223c4b022b366e99c93c85c5520d477`
`protocol_v2.json` (amendment)	`6457f447ab555e30803068fd92d7d36738d293d09616a9b4174bd0300c990549`

Table 6: Paper datasets (protocol v1). h and Δt are derived from L, T, and the grid in `protocol.json`.

PDE	grid	h	Δt	N_t	train/val/test	coefficient	boundary	data seed
Burgers 1D	32	0.1963	0.06250	16	12/4/4	$[5.00 \times 10^{-3}, 0.0500]$	periodic	0
Heat 2D	16×16	0.06667	0.04167	12	8/4/4	$[0.0500, 0.200]$	dirichlet zero	1

C The protocol amendment

The amendment `protocol_v2.json` (Table 5) makes four changes. (i) The 0 to 100score S is reported only for relL2: for residuals the best trivial baseline after excluding the zero field was the training-mean field, which made $S(\rho_{\mathrm{w}})$ strongly negative for every trained model (Table 14). (ii) Residuals are reported as the ratio C of Eq. (11). (iii) The verdict null changes from the white-noise weak-form envelope to the 95% smooth band, because by Proposition 9 the white-noise floor of ρ_{w} shrinks like $1/\sqrt{N}$, so clearing it is nearly vacuous. (iv) Model contrasts use two-sided Wilcoxon signed-rank tests over (seed, trajectory) pairs with Cohen's d_z, replacing permutation tests on seed-averaged trajectories, which left four pairs and no effect size.

D Reproducibility details

From a fresh clone of the coticula branch, `bash scripts/reproduce_paper.sh` runs `scripts/reproduce_paper.py` (protocol v1) and `scripts/reproduce_amendment.py` (protocol v2) and writes `paper_artifacts/`: CSV tables, `summary.json`, `v2_summary.json`, `manifest.json`, the two protocol files with their SHA-256 files, and the figures. In the paper source, `python3 tools/make_tables.py` reads only those files and writes `tables/*.tex` and `tables/numbers.tex`; each macro in the latter carries a comment naming its source file and row.

E Inverse case study details

Data and task. PINNForge Stage A harder Burgers pilot: periodic $x \in [-1, 1)$, $N = 1024$ spectral grid, $t \in [0, 1]$, ν uniform in $[0.005, 0.1]$, 512/128/128 train/validation/test instances. `hard_ood` is $\nu \leq 0.0270$, the training 0.25 quantile ($n = 30$ test instances). Mask: 32 equispaced sensors, bursts of 5 frames starting at frames $0, 24, 48, 72$, frame spacing 0.01. Failure: $\hat{\nu} \leq 0$ or relative error > 0.5.

Estimators. *Closed form*: least squares of the central-difference Burgers residual on the sensor grid. *Weak form*: separable test functions $(1 - \xi^2)_+^4$ in x and t, radius 8 sensor spacings, center stride 4, trapezoidal quadrature on the sensors, bursts pooled into one normal equation, no interpolation of u [Messenger and Bortz, 2021b]. *Solver, sensor lift*: the initial state is the zero-padded FFT lift of the frame-0 sensors onto the 1024-point grid; the true spectral solver is fit to the sensors of the later bursts by a 7-point coarse grid on ν plus at most 10 golden-section steps to a bracket of width 0.0005 (17.8 solves per instance on average). *Solver, oracle*: the same fit from the stored full initial field; a ceiling, not a sparse-sensor method. *FNO inverse*: a trained FNO (data-only, or hybrid with residual weight 0.01), 5 seeds, rolled out from the lifted opening burst for each of 191 values of ν; the estimate minimizes the sensor misfit on the later bursts. The clean-data FNO estimates are the committed Stage D per-instance values. The FNO wall clock is one instance of the

Table 7: Equal tuning budget: 2 trials per architecture, 40 epochs, selection on validation relL2; final models retrained for 120 epochs with 5 seeds. Sources: `protocol.json`, `summary.json`.

PDE	model	trials (config, size)	selected	val. relL2
Burgers 1D	FNO	modes=4, width=8, n_layers=2, fc_dim=32 (2049 params); modes=8, width=16, n_layers=3, fc_dim=64 (15361 params)	1	0.199
Burgers 1D	U-Net	width=8 (1217 params); width=16 (4337 params)	1	0.245
Burgers 1D	DeepONet	rank=16, hidden=32 (4577 params); rank=32, hidden=64 (15297 params)	1	0.553
Heat 2D	FNO	modes=4, width=8, n_layers=2, fc_dim=32 (9093 params); modes=8, width=16, n_layers=3, fc_dim=64 (199437 params)	1	0.256
Heat 2D	U-Net	width=8 (3317 params); width=16 (12381 params)	1	0.268
Heat 2D	DeepONet	rank=16, hidden=32 (11681 params); rank=32, hidden=64 (29505 params)	1	0.444

Table 8: Wall clock in seconds on CPU. Sources: `summary.json`, `v2_summary.json`.

run	stage	seconds
v1	noise	1.93
v1	resolution	1.55
v1	symbol	0.0291
v1	mismatch	0.0974
v1	kphi	0.843
v1	models	31.2
v1	shipped	3.19
v1	inverse	0.0500
v1	total	38.9
v2	noise bands	2.39
v2	mismatch	0.313
v2	models	29.3
v2	shipped	6.10
v2	total	38.1

191-point rollout with randomly initialized weights of the Stage D architecture (the published checkpoints are not in the repository); it measures arithmetic, not a trained network.

Stress. Table 9. Noise is added to the full field before masking. The FNO and closed-form columns are the committed Stage E scores; their FNO arms are a retrain of the Stage D architecture, so their clean cell differs slightly from Table 4.

F Additional results

Why v2 dropped the 0 to 100residual score. In v1 the residual anchor M_0 was the best trivial baseline after excluding the zero field. For ρ_w that was the training-mean field ($\rho_w = 8.43 \times 10^{-4}$ on Burgers and 2.74×10^{-3} on heat), whose residual is low because it barely changes in time and space. Every trained model then scored far below zero (for example $S(\rho_w) = -140$ for the Burgers FNO), which reflects the anchor rather than the models.

F.1 Coefficient response (secondary)

For coefficient c and $c' = \gamma c$ ($\gamma = 1.5$) we measure $S_{\text{coef}} = \|G_\theta(c') - G_\theta(c)\| \,/\, \|G^\star(c') - G^\star(c)\|$ and the cosine between the two difference fields (G_θ surrogate, $G^\star$ solver, same initial state; ideal is 1 and 1), an evaluation-only analogue of the Jacobian accuracy targeted by derivative-informed training [O'Leary-Roseberry et al., 2022, Yao et al., 2025, Yang et al., 2026]. On the v1 checkpoints (Table 15) the most accurate models respond least: $S_{\text{coef}} = 0.036$ (Burgers FNO) and 0.080 (heat FNO), with U-Net at 0.12 and 0.36. The Burgers DeepONet moves by about the right amount (1.3) in

Table 9: Noise and sparsity stress on `hard_ood` ($n = 30$): mean relative error of $\hat{\nu}$ (failures). Noise is Gaussian, as a percentage of the clean field standard deviation, noise seed 0. Weak form and sensor-lift solver from `strong_baselines_scores.json` (the solver was run only on the 4 cells shown); closed form and FNO arms from `docs/v02/stage_e_scores.json`, whose FNO arms are a retrain of the Stage D architecture (FNO failures are means over 5 seeds). One burst leaves no target for the rollout or the solver.

noise %	sensors	bursts	closed form	weak form	solver (sensor IC)	FNO data-only	FNO hybrid
0	32	4	0.349 (9)	0.0898 (0)	–	0.0510 (0)	0.0352 (0)
0.1	32	4	0.350 (9)	0.0920 (0)	–	0.0507 (0)	0.0357 (0)
0.5	32	4	0.350 (9)	0.114 (0)	–	0.0614 (0)	0.0449 (0)
1	32	4	0.349 (8)	0.148 (0)	–	0.0876 (0)	0.0692 (0)
2	32	4	0.342 (8)	0.243 (4)	0.0142 (0)	0.179 (1.8)	0.183 (2.4)
5	32	4	0.299 (8)	0.591 (15)	0.0182 (0)	0.750 (15.4)	0.835 (17.8)
0	16	4	0.653 (16)	1.81 (24)	0.154 (2)	0.175 (1)	0.161 (1)
0	8	4	2.36 (30)	82.7 (30)	2.23 (22)	2.07 (20.4)	2.03 (18.6)
0	32	2	0.317 (9)	0.108 (1)	–	0.0510 (0)	0.0402 (0)
0	32	1	0.296 (9)	0.120 (2)	–	–	–
1	16	4	0.649 (15)	2.18 (26)	–	0.209 (2)	0.200 (1.6)
1	8	4	2.38 (30)	80.0 (30)	–	2.08 (19.8)	2.03 (19)
5	8	4	2.46 (27)	75.0 (29)	–	2.66 (27.4)	2.88 (28.2)
1	32	2	0.320 (9)	0.204 (2)	–	0.0828 (0)	0.0829 (0)

Table 10: SHA-256 of the PINNForge files the inverse study depends on, as recorded in `strong_baselines_protocol.json` (the first row is recomputed from the file and matches `strong_baselines_scores.json`).

file	SHA-256
`strong_baselines_protocol.json`	`055aecf6ab7c3611c78652c8ddf727fe591158f51c670cb22f0556fddc129a86`
`stage_d_inverse_protocol.json`	`24450a3d7dec8a1c0347b099c82f27f1941f6e869f4041b320cff302adad8a0a`
`stage_d_scores.json`	`9eec8e3e177dc8f7bf4708567d3fef648f47e3fd4f9c29dce6a5e606aca614a3`
`stage_e_stress_protocol.json`	`f53f6d3d98f49335a6e64108048f74d7353b8df22543995851a07d6e428fed19`
`stage_e_scores.json`	`bb9d260b725085823992bbed47a60b5424b00f7b45a0a1b888d9f3895564e3a3`
`pilot_protocol.json`	`bc640feab6220d8871cfcdc72997bdb7cd717c1ead67d70bd347eed19ddf8c6c`

an unrelated direction (cosine -0.056); the heat DeepONet is closest to the solver (0.85, cosine 0.71). None of this is visible in relL2 or in either residual. For coefficients near the top of the training range, γc is an extrapolation.

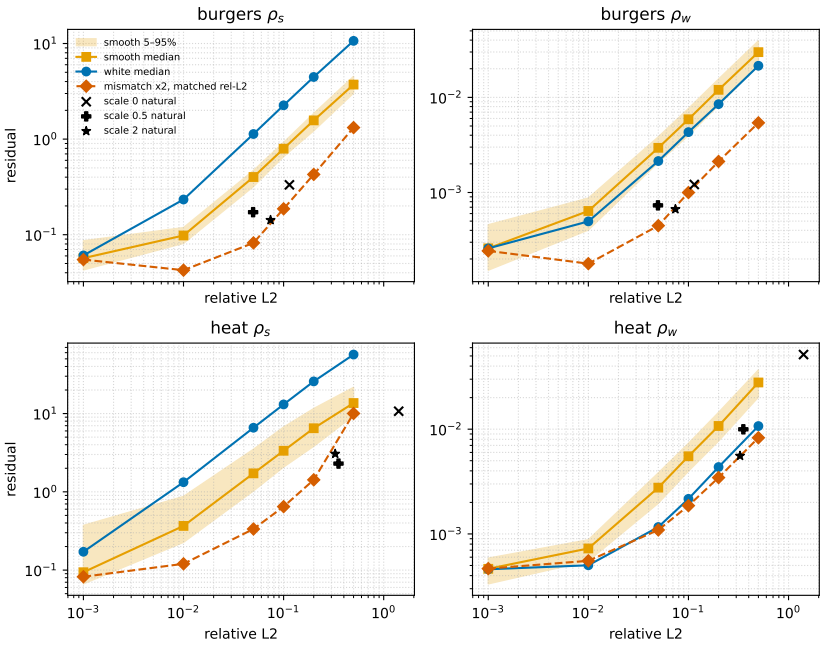


Figure 1: Residuals of exact solutions plus white, smooth (median and 5% to 95% band), and amplitude-matched coefficient-mismatch errors at matched relL2, and classical solutions at natural coefficient scales. Generated by the coticula reproduction script (`fig_matched_ladder.pdf`).

Table 11: Resolution and time-step sweep at $\varepsilon = 0.0100$, mean over seeds. Theory: second-moment formula (Proposition 3; Burgers: Proposition 8) with the per-step normalization, ignoring the discretization floor. Source: `resolution_sweep.csv`.

PDE	sweep	N	Δt	r	ρ_s white	theory	ratio	ρ_s smooth	ρ_w white	ρ_w smooth
Burgers 1D	spatial	32	0.0500	0.0440	0.283	0.284	0.996	0.0892	4.32×10^{-4}	6.28×10^{-4}
Burgers 1D	spatial	64	0.0500	0.131	0.295	0.288	1.02	0.111	3.20×10^{-4}	6.65×10^{-4}
Burgers 1D	spatial	128	0.0500	0.546	0.458	0.363	1.26	0.212	2.22×10^{-4}	5.42×10^{-4}
Burgers 1D	temporal	64	0.100	0.194	0.329	0.149	2.21	0.299	1.47×10^{-3}	1.67×10^{-3}
Burgers 1D	temporal	64	0.0500	0.131	0.295	0.288	1.02	0.111	3.20×10^{-4}	6.65×10^{-4}
Burgers 1D	temporal	64	0.0250	0.0925	0.570	0.573	0.994	0.0883	1.99×10^{-4}	6.06×10^{-4}
Heat 2D	spatial	16	0.0417	0.820	0.875	0.711	1.23	0.265	5.96×10^{-4}	7.79×10^{-4}
Heat 2D	spatial	32	0.0417	4.31	4.76	3.29	1.45	0.493	4.89×10^{-4}	7.09×10^{-4}
Heat 2D	spatial	48	0.0417	12.1	14.4	9.20	1.57	0.643	4.47×10^{-4}	7.23×10^{-4}
Heat 2D	temporal	32	0.0625	9.43	9.05	4.77	1.90	0.989	0.0161	0.0162
Heat 2D	temporal	32	0.0417	4.31	4.76	3.29	1.45	0.493	4.89×10^{-4}	7.09×10^{-4}
Heat 2D	temporal	32	0.0208	1.96	3.79	3.05	1.24	0.281	7.92×10^{-5}	5.51×10^{-4}

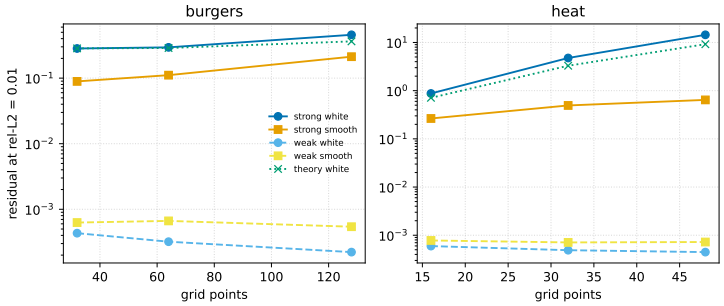


Figure 2: Residuals at relL2 = 0.01 versus grid points per side, with the second-moment prediction for white error (`fig_resolution_sweep.pdf`).

Table 12: Noise injection on exact solver outputs (protocol v1), mean ± std over 5 noise seeds of the per-seed trajectory mean. Realized relL2 equals the target to machine precision. Source: `noise_injection.csv`.

PDE	error	ε	$\kappa_h(e)\,h$	ρ_{s}	ρ_{w}
Burgers 1D	white	1.00×10^{-3}	1.56	0.0636 ± 0.0001	$2.58 \times 10^{-4} \pm 2.0 \times 10^{-6}$
Burgers 1D	smooth	1.00×10^{-3}	0.585	$0.0598 \pm 4 \times 10^{-5}$	$2.62 \times 10^{-4} \pm 2.4 \times 10^{-6}$
Burgers 1D	white	0.0100	1.56	0.234 ± 0.001	$5.12 \times 10^{-4} \pm 2.6 \times 10^{-5}$
Burgers 1D	smooth	0.0100	0.584	0.0999 ± 0.0010	$6.61 \times 10^{-4} \pm 3.0 \times 10^{-5}$
Burgers 1D	white	0.0500	1.56	$1.13 \pm 4 \times 10^{-3}$	$2.12 \times 10^{-3} \pm 4.9 \times 10^{-5}$
Burgers 1D	smooth	0.0500	0.582	0.401 ± 0.009	$3.00 \times 10^{-3} \pm 1.2 \times 10^{-4}$
Burgers 1D	white	0.100	1.57	2.25 ± 0.01	$4.34 \times 10^{-3} \pm 2.0 \times 10^{-4}$
Burgers 1D	smooth	0.100	0.583	0.787 ± 0.012	$5.94 \times 10^{-3} \pm 1.9 \times 10^{-4}$
Burgers 1D	white	0.200	1.57	4.48 ± 0.02	$8.69 \times 10^{-3} \pm 4.7 \times 10^{-4}$
Burgers 1D	smooth	0.200	0.592	1.54 ± 0.05	0.0112 ± 0.0006
Burgers 1D	white	0.500	1.56	10.8 ± 0.05	0.0213 ± 0.0012
Burgers 1D	smooth	0.500	0.585	3.74 ± 0.09	0.0302 ± 0.0011
Burgers 1D	reference	0	–	0.0592	2.55×10^{-4}
Heat 2D	white	1.00×10^{-3}	2.11	$0.236 \pm 3 \times 10^{-4}$	$4.58 \times 10^{-4} \pm 1.3 \times 10^{-6}$
Heat 2D	smooth	1.00×10^{-3}	0.844	0.138 ± 0.001	$4.60 \times 10^{-4} \pm 2.0 \times 10^{-6}$
Heat 2D	white	0.0100	2.11	$1.55 \pm 4 \times 10^{-3}$	$5.09 \times 10^{-4} \pm 9.5 \times 10^{-6}$
Heat 2D	smooth	0.0100	0.843	0.453 ± 0.008	$7.15 \times 10^{-4} \pm 1.3 \times 10^{-5}$
Heat 2D	white	0.0500	2.11	7.48 ± 0.01	$1.20 \times 10^{-3} \pm 5.6 \times 10^{-5}$
Heat 2D	smooth	0.0500	0.844	1.94 ± 0.04	$2.84 \times 10^{-3} \pm 8.2 \times 10^{-5}$
Heat 2D	white	0.100	2.11	14.9 ± 0.1	$2.15 \times 10^{-3} \pm 6.1 \times 10^{-5}$
Heat 2D	smooth	0.100	0.844	3.74 ± 0.04	$5.39 \times 10^{-3} \pm 1.4 \times 10^{-4}$
Heat 2D	white	0.200	2.11	28.7 ± 0.05	$4.36 \times 10^{-3} \pm 1.6 \times 10^{-4}$
Heat 2D	smooth	0.200	0.841	7.12 ± 0.10	0.0108 ± 0.0004
Heat 2D	white	0.500	2.11	61.7 ± 0.4	0.0105 ± 0.0005
Heat 2D	smooth	0.500	0.842	14.7 ± 0.4	0.0277 ± 0.0011
Heat 2D	reference	0	–	0.118	4.56×10^{-4}

Table 13: Paired architecture comparisons (protocol v2): two-sided Wilcoxon signed-rank test over (seed, test trajectory) pairs; Cohen's d_z = mean difference / sample std of differences. Negative difference: first model lower. Source: `v2_paired_tests.csv`.

PDE	contrast	metric	mean diff.	d_z	p	n
Burgers 1D	FNO − U-Net	relL2	−0.0337	−2.45	1.9×10^{-6}	20
Burgers 1D	FNO − U-Net	ρ_{s}	−0.582	−3.11	1.9×10^{-6}	20
Burgers 1D	FNO − U-Net	ρ_{w}	-2.00×10^{-3}	−1.23	9.5×10^{-6}	20
Burgers 1D	FNO − U-Net	C_s	−0.430	−2.52	1.9×10^{-6}	20
Burgers 1D	FNO − U-Net	C_w	−0.191	−0.824	1.7×10^{-3}	20
Burgers 1D	FNO − DeepONet	relL2	−0.302	−2.59	1.9×10^{-6}	20
Burgers 1D	FNO − DeepONet	ρ_{s}	−0.456	−3.55	1.9×10^{-6}	20
Burgers 1D	FNO − DeepONet	ρ_{w}	-7.91×10^{-4}	−1.07	2.6×10^{-4}	20
Burgers 1D	FNO − DeepONet	C_s	0.431	2.39	1.9×10^{-6}	20
Burgers 1D	FNO − DeepONet	C_w	0.248	2.50	1.9×10^{-6}	20
Burgers 1D	U-Net − DeepONet	relL2	−0.269	−2.52	1.9×10^{-6}	20
Burgers 1D	U-Net − DeepONet	ρ_{s}	0.126	0.608	0.014	20
Burgers 1D	U-Net − DeepONet	ρ_{w}	1.21×10^{-3}	0.733	2.0×10^{-3}	20
Burgers 1D	U-Net − DeepONet	C_s	0.861	4.44	1.9×10^{-6}	20
Burgers 1D	U-Net − DeepONet	C_w	0.439	2.04	1.9×10^{-6}	20
Heat 2D	FNO − U-Net	relL2	0.0441	3.37	1.9×10^{-6}	20
Heat 2D	FNO − U-Net	ρ_{s}	0.280	0.448	0.083	20
Heat 2D	FNO − U-Net	ρ_{w}	1.24×10^{-4}	0.133	0.43	20
Heat 2D	FNO − U-Net	C_s	−0.218	−1.26	2.7×10^{-5}	20
Heat 2D	FNO − U-Net	C_w	−0.166	−1.31	1.9×10^{-5}	20
Heat 2D	FNO − DeepONet	relL2	−0.125	−2.36	1.9×10^{-6}	20
Heat 2D	FNO − DeepONet	ρ_{s}	−3.25	−1.14	1.9×10^{-6}	20
Heat 2D	FNO − DeepONet	ρ_{w}	-2.54×10^{-3}	−1.07	1.9×10^{-5}	20
Heat 2D	FNO − DeepONet	C_s	−0.0408	−0.107	0.90	20
Heat 2D	FNO − DeepONet	C_w	0.0736	0.374	0.14	20
Heat 2D	U-Net − DeepONet	relL2	−0.169	−3.23	1.9×10^{-6}	20
Heat 2D	U-Net − DeepONet	ρ_{s}	−3.53	−1.10	1.9×10^{-6}	20
Heat 2D	U-Net − DeepONet	ρ_{w}	-2.66×10^{-3}	−1.12	1.9×10^{-6}	20
Heat 2D	U-Net − DeepONet	C_s	0.177	0.372	0.083	20
Heat 2D	U-Net − DeepONet	C_w	0.240	1.07	5.9×10^{-4}	20

Table 14: Protocol v1 model table with 0 to 100 scores for residuals (superseded by Table 3). The zero field is excluded from the residual anchor. Source: `models.csv`.

PDE	model	relL2	S(relL2)	ρ_s	$S(\rho_s)$	ρ_w	$S(\rho_w)$
Burgers 1D	zeros	1.000	−169	0	excl.	0	excl.
	training mean	0.997	−169	1.01	−38.4	8.43×10^{-4}	0
	frozen initial	0.371	0	0.748	0	4.84×10^{-3}	−709
	FNO	0.0754 ± 0.0010	79.7	0.446 ± 0.012	43.5	$1.63 \times 10^{-3} \pm 2.3 \times 10^{-4}$	−140
	U-Net	0.108 ± 0.006	70.9	0.993 ± 0.114	−35.3	$3.42 \times 10^{-3} \pm 1.3 \times 10^{-3}$	−456
	DeepONet	0.375 ± 0.008	−1.03	0.896 ± 0.025	−21.3	$2.45 \times 10^{-3} \pm 2.9 \times 10^{-4}$	−285
	reference	0	100	0.0539	100	2.79×10^{-4}	100
Heat 2D	zeros	1.000	−100	0	excl.	0	excl.
	training mean	0.499	0	0.961	0	2.74×10^{-3}	0
	frozen initial	1.39	−178	11.5	−1219	0.0512	−2135
	FNO	0.182 ± 0.003	63.6	4.90 ± 0.27	−455	$5.40 \times 10^{-3} \pm 4.9 \times 10^{-4}$	−117
	U-Net	0.141 ± 0.009	71.8	4.50 ± 0.29	−408	$5.12 \times 10^{-3} \pm 7.1 \times 10^{-4}$	−105
	DeepONet	0.304 ± 0.022	39.2	8.58 ± 1.52	−879	$8.15 \times 10^{-3} \pm 4.6 \times 10^{-4}$	−238
	reference	0	100	0.0940	100	4.70×10^{-4}	100

Table 15: Coefficient response at $\gamma = 1.50$ (protocol v1 checkpoints, 5 seeds). S_{coef} mean $\pm$ std over seeds; cosine is the seed mean. Source: `models.csv`.

PDE	model	S_{coef}	cosine	relL2 at c	relL2 at γc
Burgers 1D	FNO	0.0355 ± 0.0255	−0.139	0.0754	0.0776
	U-Net	0.119 ± 0.049	0.304	0.108	0.101
	DeepONet	1.33 ± 0.12	−0.0561	0.375	0.370
Heat 2D	FNO	0.0800 ± 0.0478	0.165	0.182	0.274
	U-Net	0.364 ± 0.075	0.700	0.141	0.221
	DeepONet	0.846 ± 0.087	0.710	0.304	0.356

Table 16: Classical solutions at a scaled coefficient (same initial state), scored with the original coefficient. C is the mean of per-trajectory ratios inside the noise grid; outside the grid it is not computed. Source: `v2_matched_ladder.csv` (natural_scale rows).

PDE	scale	relL2	outside grid	ρ_s	C_s	ρ_w	C_w
Burgers 1D	0	0.114	no	0.332	0.425	1.21×10^{-3}	0.233
Burgers 1D	0.500	0.0496	no	0.172	0.509	7.32×10^{-4}	0.315
Burgers 1D	2.00	0.0741	no	0.142	0.247	6.68×10^{-4}	0.171
Heat 2D	0	1.41	yes	10.7	–	0.0516	–
Heat 2D	0.500	0.353	no	2.30	0.230	9.97×10^{-3}	0.503
Heat 2D	2.00	0.328	no	3.06	0.322	5.57×10^{-3}	0.307

Table 17: Exact labels scored with a scaled coefficient in the residual (scale 1 is the true coefficient). Source: `coefficient_mismatch.csv`.

PDE	scale	ρ_s	ρ_w
Burgers 1D	0	0.179 ± 0.105	$7.70 \times 10^{-4} \pm 5.2 \times 10^{-4}$
Burgers 1D	0.500	0.0892 ± 0.0412	$3.80 \times 10^{-4} \pm 2.2 \times 10^{-4}$
Burgers 1D	1.00	0.0592 ± 0.0141	$2.55 \times 10^{-4} \pm 8.8 \times 10^{-5}$
Burgers 1D	2.00	0.250 ± 0.113	$1.09 \times 10^{-3} \pm 5.1 \times 10^{-4}$
Heat 2D	0	3.92 ± 0.90	0.0124 ± 0.0020
Heat 2D	0.500	1.97 ± 0.43	$6.15 \times 10^{-3} \pm 9.0 \times 10^{-4}$
Heat 2D	1.00	0.118 ± 0.080	$4.56 \times 10^{-4} \pm 7.5 \times 10^{-5}$
Heat 2D	2.00	3.92 ± 1.02	$9.46 \times 10^{-3} \pm 7.5 \times 10^{-4}$

Table 18: Test-function bandwidth ablation for ρ_{w} at $\varepsilon = 0.0100$ (main results use $K_\phi = 3$). Source: `kphi_ablation.csv`.

PDE	K_ϕ	reference	white	smooth	frozen initial
Burgers 1D	1	4.85×10^{-5}	4.27×10^{-4}	4.82×10^{-4}	1.81×10^{-3}
Burgers 1D	2	1.33×10^{-4}	4.81×10^{-4}	5.61×10^{-4}	3.91×10^{-3}
Burgers 1D	3	2.55×10^{-4}	5.13×10^{-4}	6.63×10^{-4}	5.61×10^{-3}
Burgers 1D	4	4.32×10^{-4}	6.17×10^{-4}	7.62×10^{-4}	6.02×10^{-3}
Heat 2D	1	1.38×10^{-3}	1.37×10^{-3}	1.48×10^{-3}	0.0435
Heat 2D	2	7.15×10^{-4}	7.39×10^{-4}	8.77×10^{-4}	0.0533
Heat 2D	3	4.56×10^{-4}	5.15×10^{-4}	7.33×10^{-4}	0.0555
Heat 2D	4	3.12×10^{-4}	3.84×10^{-4}	7.48×10^{-4}	0.0547

Table 19: Synthetic check of the heat white-error second moment, Eq. (5). Source: `symbol_heat_cases.csv`.

h	Δt	α	formula	measured	ratio
0.0500	0.0500	0.100	16800	16817 ± 243	1.00102
0.0200	0.0250	0.100	6.2820×10^5	$6.2892 \times 10^5 \pm 9526$	1.00115
0.0100	0.0250	0.120	1.4403×10^7	$1.4419 \times 10^7 \pm 2.2 \times 10^5$	1.00113
0.0312	0.0417	0.0800	68261	68338 ± 1021	1.00113

Table 20: Pre-registered hypotheses (protocol v1) and outcomes. Statistic: H1 mean paired difference (sign-flip permutation test); H2/H3 measured/formula; H4 fine/coarse strong residual; H5 weak gain / strong gain; H6 weak residual at scale 0 / scale 1. Sources: `hypotheses.csv`, `protocol.json`.

id	PDE	hypothesis	supported	statistic	p
H1	Burgers	at epsilon 0.01, strong white residual exceeds strong smooth residual (paired, both PDEs)	true	0.134	5.0×10^{-4}
H1	Heat	at epsilon 0.01, strong white residual exceeds strong smooth residual (paired, both PDEs)	true	1.10	5.0×10^{-4}
H2	Heat	heat i.i.d. second moment matches the formula within 10 percent	true	1.00	–
H3	Burgers	Burgers i.i.d. second moment matches the formula within 10 percent	true	0.992	–
H4	Burgers	strong white residual grows from the coarsest to the finest spatial grid	true	1.62	–
H5	Burgers	weak white residual grows by a smaller factor than the strong residual across that grid	true	0.318	–
H4	Heat	strong white residual grows from the coarsest to the finest spatial grid	true	16.5	–
H5	Heat	weak white residual grows by a smaller factor than the strong residual across that grid	true	0.0455	–
H6	Burgers	weak residual at coefficient scale 0 exceeds the weak residual at scale 1	true	3.02	–
H6	Heat	weak residual at coefficient scale 0 exceeds the weak residual at scale 1	true	27.2	–

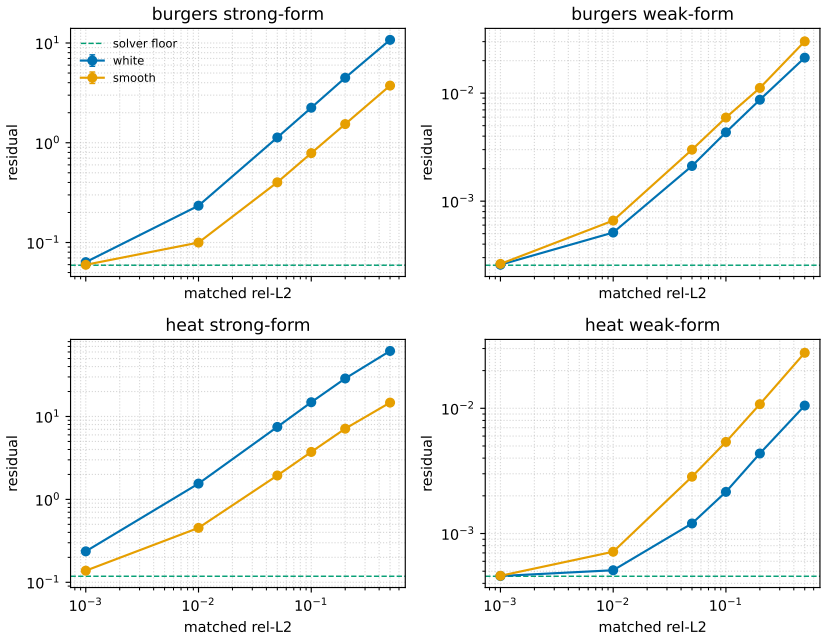


Figure 3: Strong and weak residuals versus injected relL2 for white and smooth error (protocol v1, `fig_noise_strong_vs_weak.pdf`).

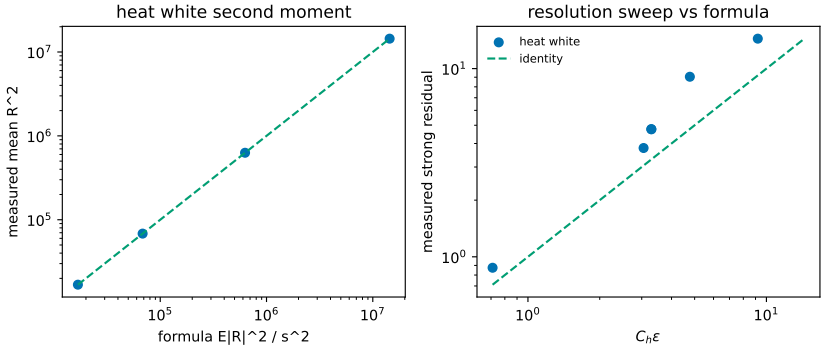


Figure 4: Measured versus predicted white-error second moments (`fig_symbol_fit.pdf`).

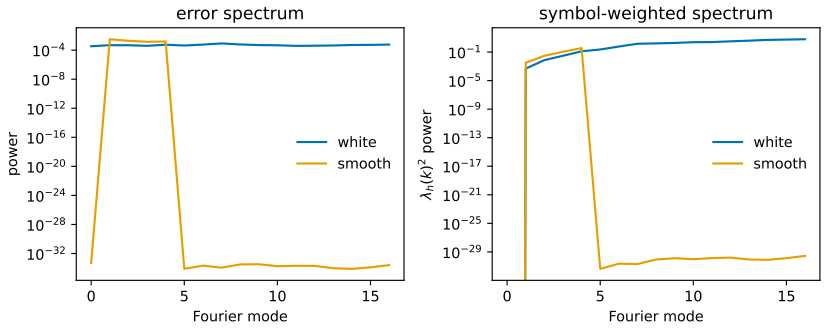


Figure 5: Error spectra of one Burgers trajectory at relL2 = 0.01, raw and weighted by the Laplacian symbol (`fig_error_spectra.pdf`).